These trading goals are: (1) minimize cost, (2) minimize cost with risk constraint, (3) minimize risk with cost constraint, (4) balance the trade-off between cost and risk, and (5) price improvement.

1. Minimize Cost

The first criterion "minimize cost" sets out to find the least-cost trading strategy. Investors may seek to find the strategy that minimizes market impact cost. If investors have an alpha or momentum expectation over the trading horizon, they will seek to minimize the combination of market impact cost and price appreciation cost. The solution of this goal is found via optimization.

$$Min\ MI' + Alpha' \tag{8.12}$$

In situations where investors do not have any alpha view or price momentum expectation the solution to this optimization will be to trade as passively as possible. That is, participate with volume over the entire designated trading horizon because temporary impact is a decreasing function and will be lowest trading over the longest possible horizon. A VWAP strategy in this case is the strategy that will minimize cost. In situations where investors have an adverse alpha expectation the cost function will achieve a global minimum. If this minimum value corresponds to a time that is less than the designated trading time the order will finish early. If the minimum value corresponds to a time that is greater than the designated trading time then the solution will again be a VWAP strategy.

Example: An investor is looking to minimize market impact and price appreciation cost. The optimal trading rate to minimize this cost is found through minimizing the following equation:

$$Min\colon b_1 \cdot I' \cdot \alpha + (1 - b_1) \cdot I' + \frac{X}{ADV} \cdot \frac{1}{\alpha} \cdot \mu \tag{8.13}$$

Solving for the optimal trading rate α^* we get:

$$\alpha^* = \sqrt{\frac{X \cdot \mu}{b_1 \cdot I \cdot ADV}} \tag{8.14}$$

Figure 8.2a illustrates a situation where investors seek to minimize market impact and alpha. Notice in this case that the efficient trading frontier decreases and then increases and has a minimum cost at \$0.16 with a corresponding timing risk of \$0.65. The trading rate that will minimize total cost is 9%. This is denoted by strategy A1 in the diagram.